

The Bond Metric: Breaking Apparent Trade-offs in High-Dimensional Embedding Geometry

A measure–metric–scale decomposition, a falsification chain,
and why one-number hubness is non-identifying

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Abstract

Summary statistics of embedding geometry — intrinsic dimension, effective rank, hubness skew, ball-growth slope — are routinely treated as properties of a corpus and are widely believed to trade off against one another. We show that several of these trade-offs are *representational artifacts*: they appear when a generative mechanism (or a learned representation) binds distinct geometric layers to a single control, and they dissolve when the layers are given separate mechanisms. We propose a four-layer decomposition of embedding geometry — global metric/spectral structure, corpus measure μ_C , query measure μ_Q , and local scale structure — and support it with a pre-registered falsification chain over a procedural corpus-generator search (seven experiment rounds against a real 1024-dimensional text-embedding corpus). Three constructive separations result: (i) angular hub structure vs. global spectrum, an apparently sharp Pareto frontier dissolved by splitting placement from colouring; (ii) *same hubness, different anatomy*: a synthetic corpus and the real corpus share the same scalar hubness reading produced by opposite mechanisms (centrality-driven super-hubs, max reverse- k NN count 369, vs. density-driven moderate hubs, max count 78) — one-number hubness is non-identifying; (iii) hubness vs. ball growth, dissolved by recognising that the real system’s hubness lives substantially in the query marginal: writing hubness as $H_k(C, Q, d)$, reverse-neighbour counts sample the query mass of top- k capture basins, $N_k(x_j) \approx |Q| \mu_Q(A_k(x_j; C, d))$. We complement the empirical chain with an *independent-realizability theorem*: the achievable hub vectors of a fixed corpus form a polytope (the convex hull of retrievable top- k incidence vectors), and an arbitrarily small simplex-address augmentation — which preserves every corpus-to-corpus neighbour ordering exactly and perturbs the spectrum by $O(\delta)$ — expands that polytope to the full hypersimplex $\Delta(n, k)$, so no universal functional law of the form $G6 = f(G2, G3, G7, G8)$ can hold over finite retrieval experiments. We conclude that apparent geometric trade-offs can be artifacts of forcing corpus metric, corpus measure, query measure, and local scale through a shared mechanism, propose a hub *anatomy vector* as the reporting standard, and derive a registered anatomy falsifier that prevents optimizers from reproducing a scalar by rebuilding a known pathology. The falsifier promptly earned its keep: in the final round the optimizer satisfied all six registered gate bands numerically while the anatomy check convicted it of rebuilding corpus super-hubs (base-to-base skew ≈ 7 , max reverse-count ≈ 350) despite having a genuine query-measure knob available — one-number geometry invites Goodharting that only anatomy-level registration catches. Pricing the anatomy into the fitness and applying one Jacobian-informed correction then closed the campaign: all six registered gates within $[0.5, 2.0] \times$ *and* anatomy in band (skew 1.4–1.9, max count 52–104 vs. real 1.5/78) on three fresh seeds — real’s numbers by real’s mechanism.

Finally we show that one-number hubness fails in a second and stronger way that is not about mechanism at all. Retrieval counts are properties of a corpus measured under a query budget, and at low occupancy the structureless expectation dominates the reading. On the real corpus the hubness of the largest grid cell is 1.27 times its own Poisson null, so a ratio gate whose null term is shared by candidate and reference is driven toward unity independently of hub structure. Three of twelve cells in the admission rule could not fail. We decompose the measured scaling slope into a null-ceiling term and a hub-structure term, finding roughly half the apparent thinning to be the ceiling, and give budget-invariant replacements including a Poisson deconvolution that recovers the attractiveness skew of the corpus itself. The same analysis yields a structural prediction that we then confirm and report as a negative: any construction whose number of attractors does not grow with the corpus fails subsample covariance, which we observe at three levels of abstraction including a fixed codebook that was expected to be safe.

1 Introduction

The setting is a benchmark-admission problem. OpenVector Bench requires a procedural corpus generator whose measured geometry matches real embeddings across eight registered gates — intrinsic dimension, ball-growth slope, effective rank, dims-to-90%, relative contrast, hubness skew, local-ID spread, and PCA retention (G1–G8) — in two measurement batteries: corpus→corpus and held-out-queries→corpus. The search for such a generator was run under a falsification discipline: before every gating round a prediction and a named falsifier were registered; every miss was reported as a miss; a sealed evaluation set was never opened. This paper is about what that discipline surfaced.

The mystery arose midway through the campaign. Successive generator families solved every gate *individually* — intrinsic dimension, rank, PCA retention, local-ID spread, hubness each fell to some construction — yet each new family hit a hard *pairwise* wall. In round 5, hubness $\times$ effective rank: twelve independent search shards (480 evaluations) mapped a clean two-ended Pareto frontier with nothing in the middle. In rounds 6–7, hubness $\times$ ball growth. Walls of this shape invite a physics reading: a genuine trade-off, a conservation law of embedding geometry, a reason the benchmark cannot be passed. The reading is wrong — and showing *why* it is wrong turns out to be more useful than passing the benchmark.

The resolution is that the walls were properties of *mechanism binding*, not of the statistics. When one generative control (a Zipf codebook hierarchy, a shared immersion, a single query recipe) sets two geometric quantities at once, those quantities trade off *under that family* — and the frontier dissolves the moment the layers are given separate mechanisms. The contributions are: **(C1)** a four-layer decomposition of embedding geometry — global metric/spectral structure, corpus measure μ_C , query measure μ_Q , and local scale structure (§2); **(C2)** the hubness-polytope and independence theorems — hubness as $H_k(\mathcal{C}, \mathcal{Q}, d)$, a linear image of the query measure, with universal approximate independence of spectrum, local scale, and query hubness by explicit construction (§3); **(C3)** three constructive empirical separations under pre-registered protocols (§5); **(C4)** the capture-basin account and the non-identifiability of scalar hubness (§6); **(C5)** the hub anatomy vector A_{hub} , the registered anatomy falsifier that caught a scalar-gaming optimizer in the wild, and a generator-specific Jacobian controllability test (§7, §3.6); **(C6)** implications for benchmark design and differential ANN testing (§9; companion paper: *Fuzzing the Bond Metric*).

Epistemic status, stated plainly: the decomposition is *strongly corroborated and materially sharpened* by the falsification chain — not universally confirmed. The theorems of §3 establish that no universal law couples the layers; whether a *particular* family or a *particular* real embedding distribution couples them is an empirical question, and §11 reports where coupling is real — including at corpus scales beyond the fitting regime, where the campaign’s own registered grid rejected its

winning candidate.

Relation to the workload-theory programme. This paper is the first output of a registered research programme (*Vector Workload Normal Forms*, `spec/NORMAL FORMS.md` in the benchmark repository) whose object of study is the joint workload $W = (P_X, P_Q, d, k, N)$ and whose target is a compact descriptor $\Phi(W)$ predicting ANN system behaviour. In the programme’s terms, the four-layer decomposition is the coarse structure of the proposed workload normal form; the theorems here are its WP1 identifiability limits (the strongest available “identical corpus diagnostics, different workload” construction); and the generator campaign is its WP3 intervention laboratory. Everything in this paper concerns geometry and measurement — whether matched geometry *plus anatomy* transfers to matched recall–cost behaviour on real ANN indexes is the programme’s next registered gate (WP5), pre-registered at the time of writing.

2 The four-layer decomposition

Layer	Main observable	Control
Global metric / spectral	spectrum, effective rank, PCA retention	recolouring
Corpus measure μ_C	cluster mass, density, angular capture	centre placement
Query measure μ_Q	visitation frequency of corpus regions	query marginal
Local scale structure	ball growth, neighbourhood transitions	radius/scale spectrum

Table 1: The Bond decomposition. “Measure” is not one thing: two interacting measures, μ_C and μ_Q , act on the same metric substrate. The query measure is an *illumination pattern* over the corpus’s capture basins.

The four layers are distinct in what they quantify over. The *global metric/spectral layer* is a property of the ambient inner-product structure after centering: the covariance spectrum and everything monotone in it. The *corpus measure* μ_C is a distribution over that substrate: where corpus mass sits, how it clusters, what angular regions it occupies. The *query measure* μ_Q is a second, independent distribution over the same substrate — operationally an *illumination pattern* over the corpus’s capture basins (§6), and the layer that six rounds of corpus-side search implicitly ignored. The *local scale structure* is the within- and across-neighbourhood distance distribution: how the first k neighbour distances of a point relate to each other and how homogeneous that profile is across query points. None of these is reducible to another: the theorems of §3 exhibit corpora in which any one is moved while the others are pinned.

Table 2 records which registered gates read which layers. The single-layer gates (G3, G4) were solved early and stayed solved; every “wall” in the campaign involved a *mixture* gate whose generative control also set another layer.

Two general principles complete the frame. *The binding thesis:* a trade-off observed under one generative family is evidence about that family’s control structure, not about the statistics — treating a family’s Pareto frontier as a law of geometry is a category error, and §5 shows the same frontier being erected and dissolved by moving one mechanism. *Amplification:* in local dimension d_ℓ , measure-layer observables respond as s^{-d_ℓ} to metric-layer perturbations of relative scale s , so binding layers is easy to do by accident and expensive to undo — tiny metric nudges produce large measure signatures, which is why an optimizer given a bound mechanism will find grotesque metric configurations to satisfy a measure target (§7).

Gate	Statistic	Layer(s) read	Registered battery
G1	two-NN intrinsic dim	local scale $\times \mu_C$ (short-range ladder)	A, B
G2	ball-growth slope	local scale (<i>across-query</i> homogeneity)	A, B
G3	effective rank	global spectral	—
G4	dims-to-90%	global spectral	—
G5	relative contrast	metric $\times$ scale mixture (weak separator)	A, B
G6	hubness skew	μ_Q over capture basins (B); μ_C anatomy (A)	A, B
G7	local-ID IQR	$\mu_C \times$ local scale mixture	A, B
G8	PCA retention	global spectral $\times \mu_C$	A, B

Table 2: Gate-to-layer map. G6 is the sharpest case: the *same scalar* read in battery B is dominated by the query measure, and in battery A by corpus anatomy — the campaign’s central confusion until round 7. G2 is an across-query homogeneity statistic (§5.3), not a within-neighbourhood one, which is why six within-neighbourhood mechanisms failed to move it.

3 Theory: the hubness polytope and the Bond independence theorem

The clean result available here is an *independent-realizability theorem*, not statistical independence. Define a retrieval experiment as the tuple

$$\mathcal{E} = (X, \mu_C, \nu_Q, d, k),$$

where $X = \{x_1, \dots, x_n\}$ is the corpus, μ_C its measure, ν_Q the query measure, d the metric, and k the retrieval depth. Classical hubness uses reverse- k NN occurrence counts [1]; order- k Voronoi cells partition query space by which k corpus points are retrieved. Throughout, assume queries in general position (ties ν_Q -null).

3.1 The hubness polytope

For $|S| = k$ define the order- k cell $V_S^{(k)} = \{q : \text{NN}_k(q; X) = S\}$ and the normalized query-induced hub mass $h_i(\nu_Q) = \nu_Q\{q : i \in \text{NN}_k(q; X)\}$. Every query contributes exactly k selections, so $0 \leq h_i \leq 1$ and $\sum_i h_i = k$.

Theorem 1 (Hubness polytope). *For any fixed corpus and metric, the set of achievable hub vectors is*

$$\mathcal{H}_k(X, d) = \text{conv}\{\mathbf{1}_S : V_S^{(k)} \neq \emptyset\},$$

where $\mathbf{1}_S$ is the 0/1 incidence vector of S .

Proof. For each query let $z(q) = \mathbf{1}_{\text{NN}_k(q; X)}$; then $h(\nu_Q) = \int z(q) d\nu_Q(q)$. Since z takes only the incidence-vector values of nonempty cells, its expectation lies in their convex hull. Conversely, given $h = \sum_S p_S \mathbf{1}_S$, choose $q_S \in V_S^{(k)}$ for each nonempty cell and set $\nu_Q = \sum_S p_S \delta_{q_S}$. $\square$

Query hubness is thus a *linear image of the query measure* — not intrinsically a corpus statistic. Equation (1) is the coordinate form of this map.

3.2 The simplex-address lemma: every hub vector becomes reachable

Theorem 1 is constrained by which cells happen to be nonempty. An arbitrarily small augmentation removes even that constraint. Let $Y = \{y_1, \dots, y_n\} \subset S^{d-1}$ be any unit-normalized corpus, and

let $v_1, \dots, v_n$ be the vertices of a regular simplex in $\mathbb{R}^{n-1}$ ($v_i^\top v_i = 1$, $v_i^\top v_j = -\frac{1}{n-1}$ for $i \neq j$, $\sum_i v_i = 0$). For $0 < \delta < 1$ construct the augmented corpus

$$x_i = (\sqrt{1 - \delta^2} y_i, \delta v_i) \in S^{d+n-2}.$$

Lemma 2 (Corpus anatomy is exactly preserved). *For $i \neq j$, $x_i^\top x_j = (1 - \delta^2) y_i^\top y_j - \frac{\delta^2}{n-1}$, a strictly increasing affine map of the original cosine similarity. Hence every corpus-to-corpus neighbour ordering, the base-to-base k NN graph, base-to-base reverse-count hubness, and every rank-based local statistic are exactly preserved. For chord distance, $\|x_i - x_j\|^2 = (1 - \delta^2)\|y_i - y_j\|^2 + \frac{2n\delta^2}{n-1}$: the full pairwise-distance matrix undergoes one common monotone reparameterization.*

Lemma 3 (Every k -subset becomes retrievable). *For any S with $|S| = k$, let $u_S = \sum_{i \in S} v_i$ and $q_S = (0, u_S/\|u_S\|)$. Then $u_S^\top v_j = \frac{n-k}{n-1}$ for $j \in S$ and $u_S^\top v_j = -\frac{k}{n-1}$ for $j \notin S$, so $q_S^\top x_j$ is strictly larger for every $j \in S$ than for every $j \notin S$, i.e. $\text{NN}_k(q_S; X_\delta) = S$. Every k -subset has a nonempty order- k cell.*

Corollary 4 (Universal query-hubness control). *For the augmented corpus X_δ , the hubness polytope is the full hypersimplex [20]:*

$$\mathcal{H}_k(X_\delta) = \Delta(n, k) = \left\{ h \in [0, 1]^n : \sum_i h_i = k \right\}.$$

Every mathematically feasible query-induced hub vector is realizable exactly (rational points exactly by finite query sets; arbitrary points to arbitrary precision), while the complete corpus-to-corpus neighbour graph is unchanged.

3.3 Spectral effect of the address block

The address block's spectral footprint is controlled, though — one correction to the naive account — not exactly diagonal. Writing the covariance of X_δ in (y, v) blocks: the diagonal blocks are $(1 - \delta^2) \text{Cov}(Y)$ and $\frac{\delta^2}{n-1} I_{n-1}$ (the regular simplex satisfies $\frac{1}{n} \sum_i v_i v_i^\top = \frac{1}{n-1} I$), so *modulo the cross term* the padded eigenvalues obey

$$\lambda_i(X_\delta) = (1 - \delta^2) \lambda_i(Y) + \frac{\delta^2}{n-1} \quad \text{— an explicit infinitesimal recolouring toward isotropy.}$$

The cross-covariance $\delta \sqrt{1 - \delta^2} M$ with $M = \frac{1}{n} \sum_i y_i v_i^\top = \frac{1}{n} Y^\top V$ is generically nonzero; by Weyl's inequality it perturbs each eigenvalue by at most $2\delta \sqrt{1 - \delta^2} \|M\|_2$, and the bound is *deterministic*: $\|Y\|_2 \leq \sqrt{n}$ (unit rows) and $\|V\|_2 = \sqrt{n/(n-1)}$ (since $V^\top V = \frac{n}{n-1} I$), so

$$\|M\|_2 \leq \frac{1}{n} \|Y\|_2 \|V\|_2 = \frac{1}{\sqrt{n-1}} \quad \text{for every unit-normalized corpus.}$$

All spectral effects vanish as $\delta \rightarrow 0$, which is what Theorem 7 uses; no genericity assumption is needed anywhere in the construction. The separation is therefore unusually strong: query hubness moves by $O(1)$; corpus neighbour anatomy moves by exactly zero; spectral change is $O(\delta)$.

3.4 Macro–micro scale separation

It remains to separate the global spectrum from local ball growth. Choose macro centres $c_a \in \mathbb{R}^p$ with minimum separation $\Delta > 0$, and for each a centered local template $U_a = \{u_{a1}, \dots, u_{am_a}\} \subset \mathbb{R}^q$ with weights β_{al} , $\sum_l \beta_{al} u_{al} = 0$. Construct $y_{al} = (c_a, \epsilon u_{al})$ with weight $\alpha_a \beta_{al}$.

Lemma 5 (Spectrum and scale occupy orthogonal blocks). *The covariance is exactly block diagonal:*

$$\text{Cov}(Y_\epsilon) = \text{Cov}(C) \oplus \epsilon^2 \sum_a \alpha_a \text{Cov}(U_a),$$

(the per-cluster centering kills the cross term identically). The $O(1)$ global spectrum is controlled by the macro centres; the microstructure enters at $O(\epsilon^2)$. At radius $\epsilon r < \Delta$ no other cluster is visible, and $\mu_C(B(y_{a\ell}, \epsilon r)) = \alpha_a \sum_{j: \|u_{aj} - u_{a\ell}\| \leq r} \beta_{aj}$: the local ball-growth curve is exactly the template's, rescaled in radius — precisely the degree of freedom behind ball-growth intrinsic-dimension estimators [6, 7].

Lemma 6 (Common-sphere realization — the bridge to the address block). *The macro-micro corpus can be constructed with all points on a common sphere, so that Lemma 2's unit-norm hypothesis is met after one global rescaling: (i) equal-norm macro centres: for any target covariance Σ with m centres there is a mean-zero ensemble with $\|c_a\| \equiv R$, $R^2 = \text{tr } \Sigma$ — take A with $AA^\top = \Sigma$ whose Gram matrix has constant diagonal (such A exists by Schur–Horn: the constant diagonal $\text{tr } \Sigma/m$ is majorized by every spectrum), then $\pm$ -double the columns for mean zero; (ii) spherical local templates: require $\|u_{a\ell}\| \equiv c$, with $\pm$ pairs on the template sphere preserving the weighted-mean-zero condition. Registered ball-growth curves remain realizable — a growth curve constrains pairwise chordal distances, which points on S^q take freely in sufficient dimension. Then $\|y_{a\ell}\|^2 = R^2 + \epsilon^2 c^2$ is constant, with no compensator coordinate (a compensator would perturb local distances at $O(\epsilon)$ — the same order as the structure being protected — and is therefore rejected).*

Without Lemma 6 the composition has a gap: the raw macro-micro corpus has unequal norms, under which Euclidean address selection can fail (norm variation is $O(1)$ while the address margin is $O(\delta)$) and cosine normalization would disturb the exact block structure. One remark survives even without it: under *cosine* retrieval the address selection of Lemma 3 is norm-robust as stated — every $j \in S$ has strictly positive inner product with q_S and every $j \notin S$ strictly negative, so $\text{NN}_k(q_S) = S$ for arbitrary corpus norms; equal norms are needed for Lemma 2's exact anatomy preservation and for Euclidean retrieval.

3.5 The Bond independence theorem

Theorem 7 (Universal approximate independence of spectrum, local scale, and query hubness). *Fix (i) a realizable target covariance spectrum Λ ; (ii) finite target local ball-growth templates; (iii) a retrieval depth $1 \leq k < n$; (iv) any feasible normalized hub vector $h \in \Delta(n, k)$; and (v) any tolerance $\eta > 0$. Then in sufficiently high finite dimension there exist a corpus X , corpus measure μ_C , query measure ν_Q , and Euclidean or cosine metric such that: $\|\lambda(\text{Cov } X) - \Lambda\| < \eta$; the registered local ball-growth profiles differ from their targets by less than η ; $h_i = \nu_Q\{q : i \in \text{NN}_k(q; X)\}$ holds exactly; and ν_Q can be varied throughout $\Delta(n, k)$ without modifying the corpus or any corpus-only observable.*

Proof construction. (1) Equal-norm macro centres realize Λ (Lemmas 5–6(i)). (2) ϵ -scale centered spherical microtemplates realize the local growth curves (Lemma 6(ii)); the corpus now lies on a common sphere — rescale to unit norm. (3) Choose ϵ small enough that the microstructure moves the global spectrum by less than $\eta/2$. (4) Append the δ -scale regular-simplex address block (Lemmas 2–3), applicable verbatim to the unit-normalized corpus. (5) Choose $\delta \ll \epsilon$ so its spectral and radial effects (the deterministic Weyl cross-term bound $2\delta/\sqrt{n-1}$ included) are below $\eta/2$. (6) Write h as a convex combination of k -subset incidence vectors (Corollary 4). (7) Place the corresponding query mass on the simplex queries q_S . $\square$

What this proves. There is no universal functional law $G6 = f(G2, G3, G7, G8)$ over all finite retrieval experiments: G6 query hubness is controlled by the query measure over top- k capture cells; G3/G8 by the macro covariance; G2 by microstructure and radius distribution; and base-to-base hub anatomy can remain *exactly* fixed while query-to-base hubness moves arbitrarily. This is stronger than the empirical observation that our knobs happened to separate.

What it does not prove. The construction is existential and deliberately permissive: the address block may require $n-1$ extra dimensions; the query and corpus marginals may differ substantially; independence is not established under $\nu_Q = \mu_C$, nor in fixed dimension; nothing guarantees the constructed corpora resemble natural embeddings; and semantic, encoder, normalization, and shared-support restrictions can reintroduce coupling. The proper claim: *there is no universal geometric necessity coupling these quantities — any coupling observed in a restricted generator or a real embedding family arises from additional structural constraints.* That is exactly what rounds 5–7 exhibit empirically.

3.6 A generator-specific controllability test (registered)

The universal theorem says nothing about whether independence holds *inside a realistic family*. For the round-7 generator define the intervention map $F : (\theta_s, \theta_q, \theta_r) \mapsto (G3, G6, G2)$, where $\theta_s = \text{`spectrum\_decay`}$, $\theta_q = \text{`query\_tail`}$, and $\theta_r =$ the local-scale-homogeneity control (`equalize`; in this family G2’s empirical controller is window homogenization rather than a radius spectrum, a fidelity note the experiment records). If the Jacobian

$$J = \begin{pmatrix} \partial G3/\partial \theta_s & \partial G3/\partial \theta_q & \partial G3/\partial \theta_r \\ \partial G6/\partial \theta_s & \partial G6/\partial \theta_q & \partial G6/\partial \theta_r \\ \partial G2/\partial \theta_s & \partial G2/\partial \theta_q & \partial G2/\partial \theta_r \end{pmatrix}$$

has full rank at the round-7 solution, the inverse-function theorem gives *local independent controllability* within the actual family. Pre-registered expectations: θ_s dominates the G3 column, θ_q the G6 column, θ_r the G2 column; J nonsingular with acceptable condition number, stable across fresh seeds.

Result (central differences $\times$ 3 seeds, at the round-7 anatomy-clean point). Mean Jacobian, rows (G3, G6, G2):

$$\bar{J} = \begin{pmatrix} -9.94 & -0.0067 & -0.0012 \\ -0.074 & 0.048 & -0.0030 \\ 0.44 & -0.011 & 0.0032 \end{pmatrix}, \quad \det \bar{J} \neq 0, \quad \kappa(\bar{J}) \approx 4.1 \times 10^3.$$

The honest reading, expectation by expectation: $\theta_s \rightarrow G3$ is *proven locally* (-9.9 , two orders dominant in its column) and was validated *prospectively*: the follow-up round’s warm-start $\Delta\theta_s = +0.15$ moved G3 $2.50\times \rightarrow 1.83\times$ (linear extrapolation predicted $1.0\times$; half the linear step, as expected on a curved response), and that warm-started point became the campaign’s solution. $\theta_q \rightarrow G6$ is *established large-signal but saturated locally*: the per-instance derivative flips sign across seeds ($+0.19/-0.12/+0.08$, same-seed central differences — landscape heterogeneity, not estimator noise) because $\theta_q = 2.35$ sits near the top of its $0.19\times \rightarrow 2.20\times$ sweep. $\theta_r \rightarrow G2$ is *locally inert* ($\sim 10^{-3}$): with `size_tail` = 0.16 the substrate is already homogeneous; G2’s largest *local* mover is in fact θ_s ($+0.44$) via the colouring’s effect on the d_{10} window. The registered condition-number expectation is **not met**: clean three-axis local controllability is *not* established at this operating point — one

axis local, one large-signal, one exhausted — and the inverse-function-theorem claim is restricted accordingly. What the campaign *does* establish is compositional: large-signal interventions on the three controls, applied in sequence across rounds 6–8, reached the joint target.

4 Experimental substrate and protocol

The real target is a 1024-dimensional Cohere Embed-V3 Wikipedia corpus with its released held-out query set; fitting used $n = 8\text{k}–16\text{k}$ working slices of the non-sealed rows (a hash-designated 25% of rows is sealed and was never loaded in any experiment reported here). Geometry is measured on the unit sphere by the benchmark’s registered battery: exact blocked k -NN, one neighbour structure per (corpus, n , subsample), gates read at $k \in \{10, 30, 100\}$, in battery A (held-out corpus rows as queries) and battery B (the real query set, or a generator’s own registered query block). Synthetic queries are held out from the *same generated instance* as their base — an independently seeded draw is a different manifold realization, and measuring against it silently penalizes every concentrated family (the query-coupling artifact; **results/** in the repository).

The search harness wraps the benchmark’s own battery in a shared `evaluate_fn` contract consumed by a 12-shard swarm random search (480 evaluations per gating round, one CPU and 2 GB per shard, on the NRP Nautilus cluster) and by a structural-fuzzing adversary. Protocol: before every gating round, a prediction and its falsifier were registered; screening smokes, when they occurred, were disclosed in the registration; misses were recorded as misses in the round documents that Table 3 summarizes.

Rd	Family / probe	Registered bet	Verdict
0	heavy-tail Gaussian clusters	loop halves mismatch	✓; G1 wall (2.4×)
1	Poincaré manifold + RFF lift	hubness via $\exp_0$	G6 0.85×; G1 worse (6.2×)
2	decoupled lift (amp)	amp frees G1	falsified ; no knob moves G1
–	elliptic curves (2 readings)	low nominal dim $\Rightarrow$ low ID	falsified (torus reads 345)
3	concentration (flat patches)	reach fixes two-NN	✓ G1+G8 solved; G6 0.15×
4	Whitney-stratified flags	dim spectrum $\Rightarrow$ G1+G7	✓ first G7 match; G6 0.18×
–	sharded lookup tables	discreteness frees G1	falsified ; G1 flat vs. knobs
5	Zipf codebook hierarchy	$G6 \geq 0.5\times$ with $G3 \geq 0.5\times$	G6 solved; joint falsified
6	+ Mahalanobis recolouring	five-gate joint point	✓ (seed-stable); G2 0.14×
7	+ query marginal (this work)	six-gate joint + anatomy	P2 fired; clean pt 5/6
8	+ anatomy-priced fitness	six gates + anatomy jointly	✓ achieved (3 seeds)

Table 3: Provenance of the falsification chain. Every “falsified” row is a pre-registered prediction that failed and was reported as a miss (**results/** in the repository).

5 Three constructive separations

5.1 Separation I: angular hubs vs. global spectrum (rounds 5–6)

- Round 5: Zipf codebook hierarchy controls both hub placement and the covariance spectrum $\Rightarrow$ clean two-ended Pareto frontier: every hub-matching shard has effective rank $\leq 0.05\times$; the one rank-matching shard sits at hubness 0.24×; nothing in the middle. Registered falsifier fired exactly as written.

- Round 6: same placement mechanism *inside* a Mahalanobis recolouring of the centred covariance (separate control) $\Rightarrow$ the frontier dissolves; first five-gate joint match (G1/G3/G6/G7/G8 all within $[0.5, 2.0] \times$, stable across 3 seeds). Recolouring *amplifies* rather than destroys the hubs.

Gate (ratio to real)	Round-5 frontier		Round-6 joint (seeds 0–2)
	hub end	rank end	
G1 intrinsic dim	1.18	0.58	0.80–0.87
G3 effective rank	0.01	1.13	0.52–0.58
G6 hubness	0.92	0.24	0.60–0.92
G7 local-ID IQR	1.06	1.24	0.94–1.04
G8 PCA retention	0.69	0.84	0.76–0.79

Table 4: Separation I. Round 5’s registered falsifier fired: no shard of 12 (480 evals) found G6 and G3 jointly above $0.5 \times$ — a clean two-ended Pareto frontier. Round 6 splits the mechanisms and the frontier dissolves; the same construction lineage holds five gates at once, stably across seeds.

5.2 Separation II: same hubness, different anatomy (the headline)

- Hub-anatomy measurement, real corpus vs. the round-6 winner at matched scalar G6: real — base $\rightarrow$ base skew ≈ 1.5 , max reverse-count 78, density-driven ($\text{corr}(N_{10}, -d_{10}) = +0.67$), hub d_{10} only 8% below population; generated — skew 7.0, max count **369**, centrality-driven, hubs 34% closer than population.
- One-number hubness is non-identifying: two systems share G6 with opposite geometry and (predictably) opposite ANN behaviour — the differential-testing demonstration is developed in the companion paper (*Fuzzing the Bond Metric*).

A_{hub} component (b $\rightarrow$ b, $k=10$)	real 8k	real 16k	Gauss null	rd-6 winner
reverse-count skew	1.50	1.36	0.33	7.00
max N_{10}	78	76	23	369
$\text{corr}(N_{10}, \text{centrality})$	+0.19	+0.12	+0.08	+0.06
$\text{corr}(N_{10}, -d_{10})$	+0.67	+0.66	+0.85	+0.12
hub vs. all median d_{10}	0.95/1.04	0.92/1.02	1.34/1.35	0.45/0.68
hub vs. all median centrality	0.52/0.48	0.52/0.48	0.02/0.01	0.92/0.75

Table 5: Separation II — same scalar G6, opposite anatomy. Real hubs are a smooth density-driven tail whose metric footprint is a mere 8% d_{10} depression inside the narrow window; the round-6 winner’s hubs are a few extreme-centrality points sitting 34% closer than population. The scalar gate cannot distinguish them; A_{hub} can.

5.3 Separation III: query measure vs. local scale (round 7)

- The G2 mechanics: with per-query neighbour distances concentrated ($d_1 \approx d_{10}$, spread $\sim 10^{1/d_\ell}$), the mean ball-count curve is $\bar{c}(r) \approx 10 F_{d_{10}}(r)$, so the fitted log–log slope over the estimator’s window (the $[0.1, 0.9]$ quantiles of d_{10}) is $\approx \log 9 / \log(\text{window})$. Real’s window is $1.17 \Rightarrow$ slope ≈ 15 ; the round-6 winner’s window is $2.14 \Rightarrow$ slope ≈ 2 . G2 is an across-query *homogeneity* statistic, not a within-neighbourhood one.

- Six corpus-side hub mechanisms fail on a metrically homogeneous substrate (scale equalization, global mean offset, Zipf duplicate families, per-point radius tilt, in-patch sub-clusters, cluster count) — all reported as the negative results they are; a per-point radius law also leaves G2 unmoved, falsifying the naive “radius spectrum” account.
- The diagnosis: real base→base skew ≈ 1.5 vs. battery-B target 6.8 — the hubness lives in ν_Q . A same-instance Zipf query marginal sweeps G6 $0.19\times \rightarrow 2.20\times$ with the window intact and G2 *improving* to $0.65\times$.
- The old G6–G2 conflict, unwound: same-distribution queries cannot produce query-induced G6 $\rightarrow$ search grows corpus-side central super-hubs $\rightarrow$ extreme density contrast $\rightarrow$ patch/plateau anatomy $\rightarrow$ G2 collapses. The conflict was between *the wrong mechanism for hubness* and realistic ball growth.
- **Round-7 verdict — the falsifier earns the paper its title.** The swarm’s best point satisfies P1 *numerically*: all six registered gates (G1, G2, G3, G6, G7, G8) within $[0.5, 2.0]\times$ on three fresh seeds (e.g. seed 0: 0.96/0.70/1.52/1.29/1.29/0.75). But the registered anatomy falsifier **P2 fired**: at that point base→base skew reads 6.8–7.8 with max reverse-count ≈ 350 — offered a genuine query-measure knob (`query_tail`), the optimizer still found it cheaper to game the scalar with corpus super-hubs (`size_tail` = 1.39, `query_tail` = 0.48). The registered anatomy check caught a scalar-gaming optimizer in the wild — the non-identifiability thesis demonstrated *by the search itself*.
- **The anatomy-filtered result.** Screening all 12 shards, exactly the three with real’s mechanism (high `query_tail`, low `size_tail`) pass the anatomy bound (skew 0.6–1.7); every wrong-anatomy shard that outscored them did so with super-hubs (up to skew 22.9, max count 1730). The best clean point (shard 1: `query_tail` = 2.35, `size_tail` = 0.16) reads G1 1.07, **G2** 1.02, G3 2.50, G6 1.36, G7 0.77, G8 0.79, with base→base skew 1.73 and max count 99 (real: 1.50/78) — five of six gates in band *with real’s anatomy*, and ball growth — pinned at $0.14\times$ through six corpus-side mechanisms — fully matched by the query model. The single miss (G3, band top 2.0) is a gate the same knob held at $0.55\times$ one round earlier; whether it is a search-budget artifact or a real coupling is exactly what the registered Jacobian test (§3.6) decides. Methodological consequence, adopted as the round-8 protocol: *anatomy belongs inside the fitness* — an unpriced anatomy is an invitation the optimizer accepts.
- **Round 8 — the registered target, achieved.** With the base→base skew scored at mandatory weight (target 1.143, measured on the real corpus with the fitness’s own estimator) and shard 0 warm-started at the Jacobian-corrected point: **all six registered gates within $[0.5, 2.0]\times$ and anatomy in band on three fresh seeds** — G1 1.00–1.05, **G2** 1.06–1.17, G3 1.83–1.92, G6 1.28–1.43, G7 0.76–0.84, G8 0.82–0.85, with b→b skew 1.38–1.91 / max count 52–104 (real: 1.50/78) — *real’s numbers by real’s mechanism*. The cold shards re-drew round 7’s candidate pools (same per-shard RNG), making the round a matched re-ranking: the anatomy-priced score promoted exactly the three query-mechanism shards plus the warm start to the top four and demoted the super-hub points — pricing flipped the optimizer’s preferred mechanism, as registered. Honest attribution: the 480 fresh evals added nothing; the winning point *is* the warm-start anchor. The campaign closed by diagnosis plus one derivative, not by search.

6 Hubness is $H_k(\mathcal{C}, \mathcal{Q}, d)$: the capture-basin account

Let $A_k(x_j; \mathcal{C}, d)$ denote the region of query space for which x_j ranks in the top k under metric d against corpus $\mathcal{C}$ — the union of the order- k Voronoi cells whose retrieved set contains j (§3). The observed reverse-neighbour count of x_j under a query workload $\mathcal{Q}$ is then, up to sampling noise,

$$N_k(x_j) \approx |\mathcal{Q}| \cdot \mu_{\mathcal{Q}}(A_k(x_j; \mathcal{C}, d)). \quad (1)$$

Corpus geometry defines the basins; the query marginal decides how heavily each is visited. The identical scalar skew can therefore arise from two opposite worlds: a few *huge basins* (corpus-side pathology — extreme central points whose basins swallow query space) or *concentrated query mass* falling on ordinary basins (the real system’s mechanism, §5.3). Equation (1) is why no corpus-only account can close the gap: it is the coordinate form of the linear map of Theorem 1, and its left factor is a property of $\mathcal{Q}$, not of $\mathcal{C}$.

This account asks for a small but consequential correction in how hubness is discussed. The classical results [1] establish hubness as a phenomenon of dimensionality with a spatial-centrality account, measured base→base — in the present notation $H_k(\mathcal{C}, d)$, a property of the corpus experiment with $\mu_{\mathcal{Q}} = \mu_{\mathcal{C}}$. That observable is real and Table 5 reports it — but it is a *different observable* from the query-measured $H_k(\mathcal{C}, \mathcal{Q}, d)$ that a retrieval system experiences, and the two need not even be of the same order (real corpus: base→base skew 1.5, query-measured skew 6.8). Hubness-reduction methods [4, 3] act on the metric layer (rescaling distances); (1) says they reshape basins, while leaving untouched the possibility that the workload’s illumination is itself concentrated. Benchmark suites whose queries are drawn from the corpus distribution [10] structurally cannot exhibit the query-induced regime — and the effect can enter through subtle channels: in this campaign’s own registered harness, battery A drew its “corpus-to-corpus” queries as the *first* held-out rows, and because Wikipedia rows arrive topically ordered, those queries were topically clustered — a de facto concentrated query marginal hiding inside the corpus battery (base→base skew for uniformly drawn queries: ≈ 1.5 ; for first-rows queries: 9.4 at $n = 25k$). The protocol was re-registered with uniform sampling once the decomposition made the asymmetry visible — the query measure is not an exotic addition to the experiment; it is present, somewhere, in every retrieval measurement, and the only choice is whether it is controlled.

7 The anatomy vector and the registered anatomy falsifier

Because a scalar underdetermines its mechanism, we propose reporting hubness as a vector:

$$A_{\text{hub}} = (\text{q} \rightarrow \text{b skew}, \text{b} \rightarrow \text{b skew}, \max N_k, \text{corr}(N_k, d_M), \text{corr}(N_k, -d_{10}), \text{top-hub mass share}),$$

where d_M is Mahalanobis centrality and d_{10} the tenth-neighbour distance. The components are cheap (they reuse the reverse- k NN counts a hubness measurement already computes) and they separate precisely the worlds the scalar conflates: Table 5’s two corpora agree in the gate and disagree in every anatomy component.

The vector’s operational form is the *registered anatomy falsifier*. In round 7 the registration required, beyond the six gate bands, that the optimizer’s best point keep base→base skew < 3 — the first pre-registered check in the campaign that looks *behind* a gate at the structure producing it. It fired (§5.3): the score-optimal point satisfied all six gate bands on three fresh seeds while manufacturing corpus super-hubs (skew 6.8–7.8, max count ≈ 350), despite having a genuine query-measure knob available. Without P2 that point would have shipped as a success. The methodological consequence, adopted as the round-8 protocol and vindicated by its result, is that

anatomy constraints belong *inside* the fitness, not only in post-hoc checks: an unpriced anatomy is an invitation the optimizer accepts.

The same move generalizes gate by gate. G1’s scalar (two-NN intrinsic dimension) deserves an anatomy separating the short-range near-duplicate ladder from manifold-bulk dimension — the real corpus’s reading is pinned by the former, a fact invisible in the scalar and decisive at scale (§11). G3’s scalar (effective rank) deserves the spectrum *shape* versus its participation-ratio summary. In each case the scalar is the trace of a mechanism, and optimizing the trace without registering the mechanism invites exactly the Goodharting round 7 exhibited.

8 Occupancy, the null ceiling, and when a hubness gate cannot fail

The preceding sections establish that a hubness scalar underdetermines the mechanism producing it. This section establishes a second and stronger sense in which the scalar fails, and it is not about mechanism at all. At low retrieval occupancy the scalar cannot discriminate between a structured corpus and an unstructured one, so a gate reading it approaches a free pass. We give the measurement, the decomposition that explains it, and the invariant forms that repair it.

8.1 Retrieval statistics are budget-relative

A cell of a scaling grid is measured with n_q queries at depth k over n points. Define the occupancy

$$\rho = \frac{n_q k}{n},$$

which is exactly the mean retrieval count per point. Any statistic compared across a varying n must either hold ρ fixed or be expressed in a form invariant to it. Raw counts and raw count maxima are neither, and a target quoted without its budget is not a target.

This is not a hypothetical. A grid that holds n_q fixed while subsampling the corpus to each n , which is the natural implementation, moves ρ by the full range of the ladder. Over a grid spanning n from 25,000 to 200,000 at $n_q = 10,000$ and $k = 10$, occupancy falls from 4.0 to 0.5. Most points at the top of that ladder are never retrieved at all.

8.2 The null ceiling moves, and it moves both statistics

Let the counts carry no structure whatever, so that each point receives a $\text{Poisson}(\rho)$ count. Two facts follow immediately and neither is small.

The busiest point still reaches a nonzero count. Writing $\text{null}_{\max}$ for the largest c such that at least one of n independent $\text{Poisson}(\rho)$ points is expected to reach c , this ceiling falls up a fixed-budget ladder, because occupancy falls faster than n grows. A raw count maximum inherits that fall whatever the corpus does.

The count vector is also skewed. A $\text{Poisson}(\rho)$ vector has skewness exactly $\rho^{-1/2}$, so the null term of a skewness statistic *rises* as occupancy falls. The two null terms therefore move in opposite directions up the same ladder, which is enough on its own to make raw readings of the two statistics disagree about the corpus while both are behaving correctly.

It is tempting to subtract the ceiling from a measured slope and call the residual the corpus contribution. Table 6 performs that subtraction on a real embedding corpus, and by that arithmetic roughly half of the apparent thinning at $k = 10$ and most of it at $k = 100$ is ceiling rather than corpus. We report the table because the arithmetic is correct and because the conclusion drawn

k	raw slope per decade	null ceiling	hub structure
10	-0.728	-0.399	-0.329
30	-0.729	-0.538	-0.191
100	-0.746	-0.635	-0.112

Table 6: Ladder slope of the real corpus, decomposed by subtracting the Poisson ceiling. The raw count maximum falls at a rate that is largely the ceiling falling with it, so a generator fitted to the raw slope is fitted in part to the arithmetic of the measurement. The residual column is *not* the corpus contribution, for the reason given in §8.5.

k	$n=25,000$	50,000	100,000	200,000
10	3.19×	2.14×	1.58×	1.27×
30	6.05×	4.01×	2.66×	1.85×
100	11.07×	7.95×	5.34×	3.54×

Table 7: Real hubness divided by the structureless expectation at the same occupancy. At the top of the ladder at $k = 10$ the real corpus is 1.27× chance, so the measured value is 79 percent null.

from it is not. Dividing or subtracting a null removes the location of the null and not its shape, and the shape of an extreme value distribution changes with occupancy. The residual therefore still carries budget. Section 8.5 measures this directly and gives the form that does survive.

8.3 A gate that cannot fail

The consequence for the admission rule is sharper than a bias. Table 7 reports the real corpus hubness as a multiple of its own null, cell by cell.

A gate compares candidate to reference as a ratio. The null term depends only on ρ , which candidate and reference share by construction, so where the null dominates both readings the ratio is driven toward unity independently of hub structure. At $k = 10$ and $n = 200,000$ the gate is not merely weak. It is close to unable to fail. In a rule that requires hubness to pass in every cell, three of twelve cells were free passes, and this was invisible for the entire campaign because the scalar reports a plausible number.

8.4 Invariant forms

Three reductions repair this and they are cheap. Write c for the count vector and $\bar{c} = \rho$ for its mean.

The *hub share* $\max_i c_i / (n_q k)$ removes the budget but keeps the ceiling. It is the right form for comparing across budgets and the wrong form for a claim about a corpus.

The *hub excess* $\max_i c_i / \text{null}_{\max}$ removes the ceiling as well. A value of one means the busiest point is no busier than chance makes it.

For the skewness route the repair is a deconvolution. Model a count as $\text{Poisson}(\rho w)$ with w the attractiveness of the point under the query measure, normalised so that $\mathbb{E}[w] = 1$. Then

$$\text{Var}(c) = \rho + \rho^2 \text{Var}(w), \quad \mu_3(c) = \rho + 3\rho^2 \text{Var}(w) + \rho^3 \mu_3(w),$$

so both moments of w are recoverable and the skewness of w is a property of the corpus and the query measure alone. On a fixed corpus measured at $\rho \in \{0.5, 1, 2, 4\}$ the raw skewness moves from 2.46 to 5.59 while this estimator holds between 9.84 and 10.02 against a true value of 9.85. An

statistic, $k = 10$	constant ρ	fixed n_q	gap
attractiveness skew	+0.543	+0.474	0.07
tail excess	+0.399	−0.513	0.91
hub excess	+1.101	−1.034	2.14

Table 8: Same rows, same code, same seeds, two budget protocols. Only the deconvolution returns the same answer. The other two reverse sign, so a claim about corpus scaling stated in them is a claim about the protocol.

intermediate repair of the form $s_k\sqrt{\rho}$ is not invariant. It overcorrects, moving from 1.74 to 11.19 on that same data, which is further than the raw statistic moves.

One caution belongs next to the estimator rather than in a later section. The deconvolution divides by $\text{Var}(w)^{3/2}$, so under a heavy tailed popularity law the third moment is carried by few points and the estimator inherits their draw noise. Measured on a codebook family with a Zipf popularity law, per-seed slope estimates spanned -1.26 to $+7.18$ while a tail-mass statistic on the same cells agreed across seeds to within five percent. The deconvolution is the high variance choice and it is nonetheless the required one, for the reason the next subsection measures.

8.5 Only the deconvolution is invariant

Removing a null by division is not the same as removing the budget. We measured a real corpus under two budget protocols on identical rows, code and seeds, one holding occupancy fixed across the ladder and one holding the query count fixed. A statistic that is invariant must return the same ladder slope under both.

The reason is that dividing by a null removes the location of the null and not its shape. The distribution of an extreme value, and of a tail mass, changes form with occupancy, so a ratio to its own expectation still moves with the budget. The moment deconvolution removes it because it models the sampling explicitly and inverts it.

Two consequences follow and both are uncomfortable. First, the residual of Table 6 is not the corpus contribution, and no subtraction of that kind can be. Second, and larger, the direction of the real corpus’s hub scaling is the reverse of what the fixed-budget reading suggests. Measured in the invariant form the attractiveness skew of the real corpus *rises* with n , at $+0.47$ to $+0.54$ per decade at $k = 10$ and $+0.37$ to $+0.45$ at $k = 30$, consistently under both protocols. Read through the raw statistics the same corpus appears to thin. A generator calibrated against the thinning reading is being asked to reproduce the opposite of the corpus’s behaviour.

8.6 Fixed owners fail at every level of abstraction

The most persistent negative result of the campaign acquires a cleaner statement in this language. Hub mass in a real corpus behaves as a population law that re-expresses at every sampling scale. A construction that carries hub mass in a fixed set of owners cannot imitate that, because its owners either keep their counts under thinning or vanish with it.

The same failure has now been met at three levels. It was first met with planted owner rows. It was met again with planted counts, which survive subsampling while the reference thins. It was met a third time with a fixed codebook, where hub mass lives in atom popularity rather than in rows. The third case was predicted to be safe, on the argument that a popularity law over a codebook is a population law and therefore scale free. Measured, its hub concentration *rises* with

corpus size where the real corpus declines, because adding rows to a codebook of fixed size crowds the same attractors. A fixed codebook is a fixed set of owners one level up.

We state this as the operative lesson rather than as a conjecture. Any construction in which the number of attractors does not grow with the corpus will fail subsample covariance, and the direction of the failure is predictable from whether the attractors are enumerable.

9 Implications

For benchmark design. A corpus-only generator cannot reproduce query-induced statistics — not as a matter of tuning but of type: the observable is a functional of μ_Q , which the generator does not model. Synthetic workloads therefore need an explicit, registered query process, sharing the corpus’s structural realization, with its own marginal. This is now the admission convention in the benchmark this campaign served.

For ANN evaluation. Same-G6 corpora with opposite anatomy are natural differential test vectors: run exact k NN against HNSW, DiskANN, FAISS, and cuVS builds; fp32 against quantized operating points; cosine against Euclidean; alternative graph build orders and shardings — with oracles on hub-rank divergence and on anti-hub-query recall collapse. Aggregate recall hides exactly the tail the decomposition exposes: a system can score identically on two corpora sharing every scalar gate and fail catastrophically on the queries that land in one corpus’s anti-hub region. The companion paper (*Fuzzing the Bond Metric*) develops the coverage signature over layer-separated cells and the differential harness.

For representation learning. Apparent statistic trade-offs in learned embeddings deserve the same suspicion the generator walls received here. An architecture, like a generator family, is a control structure binding geometric layers to shared mechanisms (normalization couples radial to angular structure; attention temperature couples spectrum to clustering); an observed trade-off across training runs is evidence about that control structure, and the constructive question — what factored mechanism would dissolve it? — is as answerable there as it was here.

10 Related work

Hubness. The phenomenon — some points appearing unusually often in k NN lists as dimension grows — was established with a spatial-centrality account by Radovanović et al. [1], following the distance-concentration literature [2]. Hubness-reduction methods — local scaling [3], mutual proximity [4] — transform distances so that reverse-neighbour counts flatten. In the present frame these are metric-layer interventions on basin shapes; the decomposition adds that the observable itself is a functional of the query measure (1), so an untransformed corpus under a concentrated workload and a pathological corpus under a uniform workload can present the same scalar — a distinction invisible to corpus-side treatment and, per Theorem 7, unresolvable from corpus-only observables. From the index side, a recent experiments-and-analysis study [14] identifies hubness as the fundamental cause of long-tail latency instability in graph ANNS — corroborating the treatment of hubness here as a first-class workload property rather than a nuisance statistic.

Difficulty measures and workload shift in ANN. Relative contrast [5] and local intrinsic dimensionality [6, 7] are corpus- or query-local difficulty accounts; Aumüller and Ceccarello [12] use LID and relative-contrast profiles to assemble query workloads of controlled difficulty, and recent graph-native measures (Steiner-hardness [8]) explicitly tie difficulty to the index structure. Out-of-distribution query work is the nearest prior art to the query-coupling protocol effect of §4:

OOD-DiskANN [9] and RoarGraph [13] document that cross-modal query/corpus mismatch degrades graph-ANN performance (RoarGraph measures OOD queries lying $2.1\text{--}11.3\times$ farther from their true neighbours), and the big-ann-benchmarks OOD track [11] made such workloads standard. The line to draw: that literature treats *genuinely different* query distributions as an engineering problem for index construction, and [12]’s same-sample query splits behave consistently across random draws — the complementary observation. None of it treats the query set as part of the *measurement operator*, whose sampling protocol (same structural realization vs. independent draw of a synthetic generator) can silently invalidate the difficulty and intrinsic-dimension readings themselves. Standard benchmark practice [10, 11] fixes real query sets but characterizes datasets by corpus statistics; the decomposition explains why that characterization underdetermines system behaviour and what to report instead.

Synthetic vector generation for ANN benchmarking. Prior generators are single-knob or uncalibrated: Elliott and Clark [19] generate vectors of prescribed intrinsic dimension by Gaussian combination within a linear subspace (one knob; effective rank tied to ID by construction), and industrial tooling synthesizes benchmark vectors as noise around a handful of sample seeds with no geometric target at all. Tabular synthesis lines (copula- and GAN-based) preserve marginals and correlation structure, not neighbourhood geometry, and we find no application of them to embedding corpora. No prior generator we could locate matches even two of the diagnostics measured here simultaneously, calibrates to measured values of a real embedding corpus, or exposes hubness as an independent control at fixed dimension and intrinsic dimension — hubness in prior synthetic data is always emergent, dialable only implicitly through dimension [1]. The campaign reported here is, to our knowledge, the first to attempt — and the registered admission the first to gate — joint calibrated matching.

Intrinsic-dimension estimation. TwoNN [7] and maximum-likelihood [6] estimators read short-range distance ratios; their sensitivity to near-duplicates and sampling density is exactly the local-scale layer’s signature, and §11 reports it as the binding constraint at scale. Fragments of the query-coupling mechanism (§4) are separately on record: noise inflates TwoNN toward ambient [7], multiscale SVD exhibits the noisy-manifold crossover from intrinsic to ambient readings [18], LID is defined from the outset as a query-relative quantity [15], and off-manifold probes — adversarial or out-of-distribution points — read anomalously high LID [16, 17]. We find no prior statement, however, that probing an estimator with queries from an *independent realization* of the same synthetic generator reads ambient while same-realization holdouts read the true dimension — i.e. that for synthetic workloads the query protocol is part of the estimator.

Fuzzing and property-based testing. The registered-falsifier discipline and the companion paper’s coverage signature transfer methodology from coverage-guided fuzzing [22] and property-based testing [21]: treat a corpus generator as a program under test, geometric cells as coverage, and gate-satisfying-but-anatomy-violating corpora as the crashing inputs.

11 Limitations and the road to a theorem

Corroboration, not universal confirmation. The empirical chain is three separations on one real corpus family (text embeddings, one model class) and one generator lineage. The theorems are existential: they forbid universal coupling laws but do not predict which couplings a particular family exhibits.

The scale postscript — reported as the miss it is. After the campaign closed at its fitting scale ($n = 8k$), the benchmark’s registered grid admission (n from 25k to 200k, $\pm 15\%$ equivalence bands, scaling- exponent criteria) rejected the round-8 candidate: 0 of 24 cells. The rejection

map is itself decomposition-shaped, and each miss lands in a nameable layer. The G3/G4 miss is flat in n — a mis-tuned colouring (metric layer; the knob exists). G1 drifts upward with n (+0.04–0.06 per decade) where the real corpus is n -flat — real’s two-NN reading is pinned by a graded short-range ladder (nearest-neighbour distances at *fractional* scales: 1%-quantile 0.375 vs. median 0.86) that uniform-sparse patches lack; this is the G1 anatomy of §7, now load-bearing. And every scaling-exponent failure is G6 growth: a fixed query concentration gives +0.13/decade where the real workload gives +0.22 — capture basins shrink as $\sim 1/n$ while real’s per-basin query mass does not, so μ_Q is n -coupled. Both frontier mechanisms (the graded ladder; the n -coupled illumination) entered the campaign as predictions of the decomposition rather than post-hoc repairs; their registered test is the current round.

Scope of the query-coupling effect. The ambient two-NN reading under independent-seed queries (§4) is a finite-sample, off-the-empirical-support phenomenon, and the claim is scoped accordingly. “Independent realization” means a *re-drawn structural realization*: the generators here re-draw the manifold itself — patch centres, cluster hierarchy, colouring — with the seed, so an independently seeded query lands off the corpus realization’s support, its first and second neighbour distances nearly coincide, and the ratio estimator reads ambient. For query points re-drawn on a fixed, genuinely shared continuous manifold, NN-radius theory predicts the intrinsic reading, and the effect must vanish as sampling density grows. The finding is therefore a statement about measurement protocol for finite synthetic workloads — the query process is part of the estimator and must share the corpus’s structural realization — not a property of manifolds; stated without this scoping it would be refutable.

What a theorem needs. A formal generative class in which the four layers are independently parameterized, with proofs that each gate is invariant to the other layers’ controls within stated ranges; candidate assumptions: unit-sphere metric, sub-Gaussian patches, capture-basin regularity, query support contained in the corpus’s structural realization.

Couplings that are real, not representational. The s^{-d_ϵ} amplification is a genuine cross-layer sensitivity, not an artifact; normalization couples radial to angular structure; and encoder families may impose joint constraints a factored generator does not. The decomposition should be expected to fail exactly where such shared physical mechanisms exist — finding them is what the framework is *for*: a coupling that survives mechanism-splitting is a discovery about the system, not about the statistics.

12 Conclusion

Six rounds were spent solving a corpus-side problem whose target lived substantially in the query marginal; when the right knob finally existed, the optimizer preferred to rebuild the old pathology, and only a falsifier that looked behind the scalar caught it; pricing that anatomy into the objective and taking one derivative-informed step then reached, in a single round, the joint target eight rounds of scalar-gate search could not — real’s numbers by real’s mechanism, stable across seeds. The statistics were never in conflict; the mechanisms were. The registered scale grid then rejected the same candidate beyond its fitting regime (§11) — and each miss landed in a nameable layer, with the two frontier mechanisms entering as predictions of the decomposition rather than repairs. Embedding geometry should be reported as an experiment (X, μ_C, ν_Q, d, k) , its hubness as an anatomy, and its trade-offs as claims about control structure — claims that a small amount of construction, or one pre-registered look behind a gate, can refute.

Reproducibility. All rounds, pre-registrations, falsifier verdicts, and raw numbers: <https://github.com/ahb-sjsu/openvector-bench> (results/, spec/BOND_METRIC.md). Sealed evaluation rows were never loaded during any experiment reported here.

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